

Ethan Lee

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EDUCATION

University of California, Los Angeles

Bachelor of Science in Computer Science

Los Angeles, CA

Sep. 2023 – June 2027

- **Relevant Coursework:** Data Structures & Algorithms, Computer Organization, Software Construction, Machine Learning, Operating Systems, Computer Graphics, Linear Algebra, Discrete Math

EXPERIENCE

Capital One

Software Engineer Intern

Richmond, VA

Jun. 2026 – Aug. 2026

- Enabled 8 beta teams to configure custom spend-threshold alerts routed via email/Slack instead of manual checks.
- Cut cost-anomaly detection time from 2 days to under 1 hour during an 8-team beta pilot.
- Automated cost anomaly detection w/ EventBridge and CloudWatch, replacing 5 hrs/week of manual monitoring.
- Built serverless Go REST API on AWS Lambda with Angular/TypeScript dashboard for cross-service alerts.

PROJECTS

Reflectory | *Next.js, TypeScript, Supabase/Postgres, pgvector, Gemini API*

Aug. 2026 – Sep. 2026

- Engineered a RAG pipeline with pgvector and Gemini to ground AI reflections in users' own journal history.
- Designed a two-layer crisis-detection system (keyword + LLM classifier) to reroute at-risk users to crisis resources.
- Enforced per-user data isolation via Supabase RLS, verified through independent multi-account testing.
- Implemented atomic Postgres RPC rate-limiting across Vercel's multi-instance runtime, backed by 112 Vitest tests.

The Primer | *React, TypeScript, Python, FastAPI, PostgreSQL, Gemini API*

Aug. 2026 – Sep. 2026

- Engineered adaptive Bayesian Knowledge Tracing engine, capping difficulty swings to 1 level per attempt.
- Decoupled deterministic grading from an LLM explanation layer so a Gemini outage degrades gracefully.
- Optimized PostgreSQL access with targeted indexes and eager-loading, eliminating an N+1 query bottleneck.
- Built a React frontend with a JWT-authenticated parent dashboard tracking per-skill mastery and streaks.

TrialFlow | *React, TypeScript, Express.js, Creao, Composio MCP, Google APIs*

Oct. 2025

- Won Regeneron Runner-Up at CalHacks 12.0 (600+ teams) with a platform tackling 40% patient dropout rates.
- Engineered a custom Burden Score API via Creao to quantify participant workload based on procedure complexity.
- Automated patient self-scheduling by integrating Google Calendar APIs for real-time appointment sync.
- Automated 24-hour notification pipelines delivering appointment logistics and personalized prep instructions.

Bruin Plan | *MongoDB, Express.js, React, Node.js, Google OAuth*

Mar. 2025 – May 2025

- Architected MERN stack app processing 15,000+ course combinations to optimize student schedules.
- Engineered a UCLA course-catalog scraper enabling students to sort schedules by proximity or time of day.
- Integrated Google OAuth 2.0 authentication, improving signup-to-login conversion from 55% to 80%.
- Developed dynamic programming algorithm reducing average schedule conflicts from 3.1 per user to 2.0.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Python, C++, HTML/CSS, SQL (PostgreSQL, MongoDB)

Frameworks: React, Next.js, Angular, Express.js, FastAPI, Node.js, Tailwind CSS, SQLAlchemy

Tools & Cloud: AWS (Lambda/RDS/EventBridge), GCP, Docker, Supabase, Vercel, Render, Git, Linux

AI & ML: Gemini API, OpenAI API, RAG (pgvector), Prompt Engineering, Bayesian Knowledge Tracing